

2's Complement Adder-Subtractor

1. Introduction

A **2's Complement Adder-Subtractor** is a combinational digital circuit capable of performing both binary addition and subtraction using the 2's complement method. The design uses shared hardware resources — primarily a set of full adders — to efficiently implement both operations by controlling a single control line.

This dual functionality makes it an essential building block for arithmetic logic units (ALUs) in CPUs and embedded processors.

2. Concept of 2's Complement

The **2's complement** of a binary number is a way to represent negative numbers in binary form. It is obtained through the following steps:

1. Take the **1's complement** (invert all bits).
2. Add 1 to the result.

To subtract B from A using 2's complement:

$$A - B = A + (2's \text{ complement of } B)$$

This allows subtraction to be performed using addition circuitry, simplifying hardware design.

3. Working Principle

The 2's complement adder-subtractor circuit includes:

- **XOR Gates:** Used to invert each bit of B conditionally, depending on the operation.
- **Full Adders:** Perform binary addition of inputs A and the modified B .
- **Control Line (SUB):** Determines the operation:
 - $SUB = 0 \Rightarrow A + B$
 - $SUB = 1 \Rightarrow A - B = A + (\sim B + 1)$

4. Bit-Level Operation

SUB (Control Bit)	Operation
0	$A + B$ (Addition)
1	$A + 2\text{'s complement of } B = A - B$ (Subtraction)

5. Block Diagram (4-bit 2's Complement Adder-Subtractor)

- Each bit of B passes through an XOR gate: $B[i] \oplus \text{SUB}$
- If $\text{SUB} = 1$, the bits of B are inverted.
- The control line SUB is also connected to the carry-in of the least significant bit (LSB) full adder to account for the +1 in the 2's complement.
- All bits are processed simultaneously by a chain of full adders.

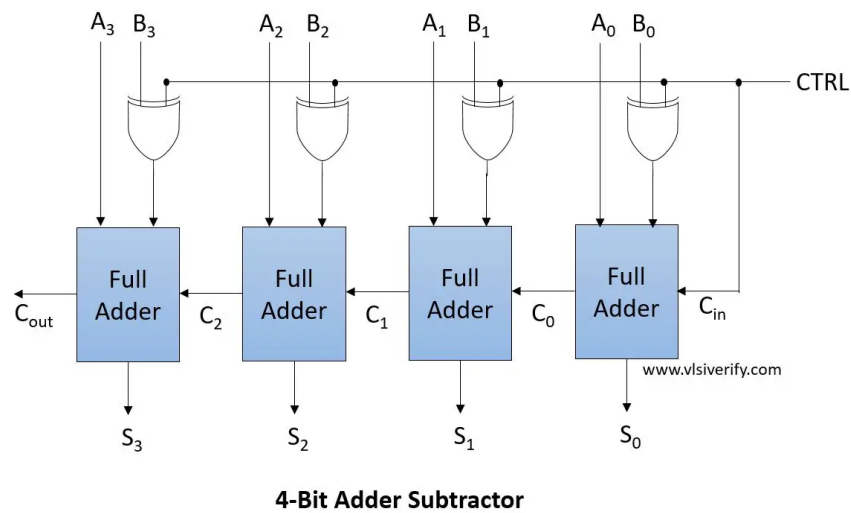


Figure 1: Block Diagram of a 4-bit Ripple Carry Adder

6. Full Adder Logic (Per Bit)

Let:

- $A[i]$: i^{th} bit of operand A
- $B[i]$: i^{th} bit of operand B
- SUB: Operation control signal

Then:

$$\text{Modified } B[i] = B[i] \oplus \text{SUB}$$

$$\text{Initial Carry-in} = \text{SUB}$$

$$\text{Sum}[i] = A[i] \oplus \text{Modified } B[i] \oplus \text{Carry}[i]$$

$$\text{Carry-out} = \text{Majority}(A[i], \text{Modified } B[i], \text{Carry}[i])$$

7. Example

Let:

$$A = 0101_2 = 5, \quad B = 0011_2 = 3$$

Addition: SUB = 0

$$\text{Result} = 0101 + 0011 = 1000_2 = 8$$

Subtraction: SUB = 1

$$B = 0011 \rightarrow 1\text{'s complement} = 1100, \quad +1 \Rightarrow 1101$$

$$\text{Result} = 0101 + 1101 = 10010_2 \rightarrow (4\text{-bit truncated}) = 0010_2 = 2$$

8. Advantages

- Performs both addition and subtraction with shared hardware.
- Requires only a control line and XOR gates to switch operation mode.
- Minimizes hardware cost and complexity.
- Used widely in ALU design.

9. Disadvantages

- Limited to 2's complement representation for subtraction.
- Overflow detection logic must be added separately for reliable results.

10. Applications

- **ALUs (Arithmetic Logic Units):** Core arithmetic operations in CPUs.
- **Digital Signal Processors (DSP):** Where arithmetic operations are frequently performed.
- **Embedded Systems:** Efficient computation in space-constrained devices.
- **Control Systems:** For difference calculation and control logic.

11. Conclusion

The 2's Complement Adder-Subtractor is an essential building block for digital arithmetic. It simplifies hardware implementation by allowing both addition and subtraction using a single adder structure, combined with XOR logic and a control line. Though it introduces challenges like overflow detection, it remains a highly efficient design for many digital systems including ALUs, processors, and embedded devices.